

Beyond the Bean: Analysis of Global Coffee Production, Quality, and Market Dynamics

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Abstract

This project uses three complementary datasets—FAO global production statistics, the Coffee Quality Institute (CQI) cupping database, and a coffeereview.com consumer review corpus—the project investigates three interconnected dimensions of the coffee industry: (1) how geography and altitude shape production yield efficiency, (2) what sensory and processing attributes drive professional quality scores, and (3) how price, roast profile, and origin interact in the specialty coffee market. The primary tools are Python (Plotly, Altair, scikit-learn, spaCy/NLTK) deployed as an interactive Plotly Dash web application. Expected deliverables include an interactive dashboard. As a application-based project, It would be both exploratory and explanatory.

Data Sets

Dataset	Description & Source	Links
FAO Coffee Production	Country-level green coffee bean production (tonnes), area harvested (ha), and yield (tonnes/ha) from 1961–2022. Sourced from FAOSTAT via Our World in Data.	Our World in Data – FAO
CQI Coffee Quality (Kaggle)	Arabica bean cupping scores from professional Q-graders: aroma, flavor, aftertaste, acidity, body, balance, uniformity, clean cup, sweetness, total cup points. Includes origin country, altitude, variety, processing method, and defects.	Kaggle – Coffee Quality Database from CQI
Coffee Reviews (Kaggle)	Consumer-facing reviews from coffeereview.com (2017–2022): blend name, roaster, roast level, origin, price per 100g (USD), rating (0–100), and three free-text description fields containing flavor notes.	Kaggle – Coffee Reviews Dataset

Tools

The project will be built entirely in Python. Core visualization libraries include Plotly (for interactive charts, choropleth maps, and radar/polar plots) and Altair (for linked, declarative small-multiple views). The dashboard framework will be Plotly Dash. Data wrangling will use Pandas and NumPy. Natural language processing on review descriptions will use NLTK or spaCy.

Visualization Process

The project is structured around four hypotheses:

H1. Geography & Production yield: Countries near the equatorial "coffee belt" (23.5°N–23.5°S) achieve higher yields (tonnes/ha) than those at the margins

Visualizations: Animated choropleth map of production over time, yield-vs-latitude scatter plot with trend lines.

H2. Quality drivers: Altitude is the strongest single predictor of total cup points among geographic/processing features—beans grown above 1,500m score significantly higher than those below 1,000m.

Visualizations: Altitude-binned box plots of total cup points, interactive parallel coordinates across 10 cupping dimensions (filterable by country, variety, processing method), and a regression coefficient chart for quality predictors.

H3. The price–rating relationship is non-linear: spending above ~\$12/100g yields diminishing quality returns.

H4. NLP analysis of review descriptions will reveal distinct flavor-descriptor clusters (e.g., floral/fruity vs. chocolatey/nutty) that correspond to origin geography.

Visualizations: Price-vs-rating scatter, word cloud by rating tier, flavor-descriptor network, and roast-level distribution by year.

The three datasets will be linked at the country level to ask: do high-volume producers also produce high-quality beans, or is there a volume–quality tradeoff? Visualized as a bubble chart (production volume vs. cupping score, sized by price).

Expected Deliverables

- Interactive Plotly Dash Web Application
- Analytical Write-Up: A comprehensive report documenting methodology, hypothesis testing results, key insights, and design rationale for each visualization. Includes discussion of limitations and what the data cannot tell us.
- Reproducible GitHub Repository